

**MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION
TECHNOLOGY
FACULTY OF SOFTWARE ENGINEERING**



**SOFTWARE DEVELOPMENT THESIS PROJECT
COURSE: CT250H**

Thesis title

**DESIGN AND IMPLEMENTATION OF A HUMAN-IN-THE-LOOP
AI-ASSISTED RECRUITMENT AUTOMATION PLATFORM
FOR CV-JD EVALUATION AND RECOMMENDATION IN IT**

Student

Pham Huu Hung - B2203557

Can Tho, August 2026

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**Supervisor
Ph.D. Nguyen Thanh Khoa**

**Student
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Can Tho, August 2026
Pham Huu Hung

DECLARATION OF ORIGINALITY

I hereby declare that this thesis, titled “Design and Implementation of a Human-in-the-Loop AI-Assisted Recruitment Automation Platform for CV-JD Evaluation and Recommendation in IT,” is my original work completed under the supervision of Ph.D. Nguyen Thanh Khoa. The implementation, experiments, data analysis, and conclusions presented in this report are reported truthfully within the stated scope.

All ideas, publications, standards, software documentation, figures, and other materials obtained from external sources are acknowledged through citations or source notes. I accept full responsibility for the accuracy, academic integrity, and originality of the submitted work.

Can Tho, August 2026
Student

Pham Huu Hung

SUPERVISOR'S COMMENTS

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Can Tho, August 2026
Supervisor

Ph.D. Nguyen Thanh Khoa

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LIST OF ABBREVIATIONS

Abbreviation	Full Term	Meaning / Explanation
AI	Artificial Intelligence	Computer systems designed to perform tasks commonly associated with human intelligence
API	Application Programming Interface	A defined interface through which software components communicate
CV	Curriculum Vitae	A document describing a candidate's qualifications and experience
E2E	End-to-End	Testing of a complete workflow across integrated components
HITL	Human-in-the-Loop	Automation in which humans retain oversight or approval authority
JD	Job Description	A description of a job's responsibilities and requirements
JWT	JSON Web Token	A compact token format used for authentication and authorization
MRR	Mean Reciprocal Rank	A ranking metric based on the position of the first relevant item
nDCG	Normalized Discounted Cumulative Gain	A ranking metric that accounts for relevance and position
NLP	Natural Language Processing	Methods for processing and analyzing human language
OCR	Optical Character Recognition	Extraction of machine-readable text from images or scanned documents
RBAC	Role-Based Access Control	Authorization based on assigned user roles
REST	Representational State Transfer	An architectural style for networked APIs
TF-IDF	Term Frequency–Inverse Document Frequency	A weighting method for representing text terms
UAT	User Acceptance Testing	Validation of whether the system satisfies intended user workflows

ABSTRACT

Background: Recruitment in the information technology sector requires candidates and recruiters to compare many types of information from curricula vitae and job descriptions. Most job portals support posting, searching, and manual applications, while automated screening tools may not clearly explain how a score is calculated or how an automated action is controlled.

Objectives: This thesis designs and implements CareerFit IT AutoPilot, a web-based recruitment platform that combines a job portal, CV-job description matching, personalized job recommendations, policy-based automation, and Human-in-the-Loop interaction in one auditable workflow.

Methods: Recruitment text is normalized and represented with Term Frequency-Inverse Document Frequency vectors. Cosine similarity ranks CV-job pairs and candidate-job recommendations, while Rocchio relevance feedback updates learned representations from explicit feedback. AutoFit checks scores, consent, thresholds, interaction history, quotas, cooldowns, time zones, and quiet hours before choosing an action. CareerFit uses a Spring Boot backend, a React and TypeScript frontend, and PostgreSQL persistence managed through Flyway migrations.

Results: The refreshed backend suite passed 72 of 72 tests, the production frontend build completed successfully, and all 20 Chromium workflow, contract, and resilience tests passed. In the controlled synthetic benchmark, nDCG@5 increased from 0.037737 to 0.837737 after Rocchio feedback. These results support a well-tested academic prototype, but they do not establish production readiness or real-world hiring effectiveness.

Keywords: recruitment automation; CV-job matching; recommendation system; Human-in-the-Loop; Rocchio feedback; explainable automation.

INTRODUCTION

Problem Statement

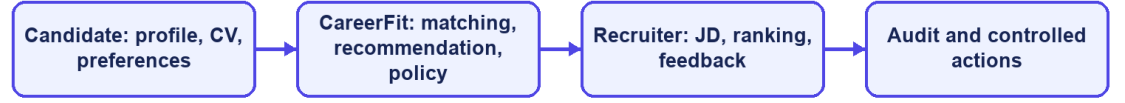
IT recruitment requires candidates and recruiters to compare many details, including technical skills, experience, language, location, and working conditions. Candidates must identify suitable vacancies, while recruiters must review CVs and decide which applicants deserve closer attention. As the number of jobs and profiles grows, doing this manually becomes slow and inconsistent [10].

Most job portals support posting, searching, and applying, but they do not fully explain why a job fits a candidate or why one applicant ranks above another. CareerFit therefore treats CV–JD matching and personalized recommendation as related but separate tasks. The first compares a specific CV and job; the second prioritizes jobs from a candidate's broader profile and preferences.

Automation can reduce repetitive work, but a similarity score should not directly trigger an important action. User consent, application state, previous interactions, thresholds, quotas, and special conditions also matter. CareerFit uses a Human-in-the-Loop design so users can review important actions and administrators can trace what happened.

This thesis develops CareerFit IT AutoPilot, an IT-focused web platform that combines job discovery, CV and JD processing, matching, recommendation, Rocchio feedback, AutoFit policies, email actions, and audit records. The aim is not to replace recruitment judgment, but to provide understandable evidence and controlled assistance throughout the workflow.

CareerFit problem context



CareerFit IT AutoPilot — thesis diagram

Figure 1.1. CareerFit problem context and principal information flows

Background and Related Work

The first motivation is to reduce fragmentation. Candidates often move between job search, CV management, application history, and email, while recruiters switch between vacancy management, applicant review, candidate discovery, and reporting. CareerFit brings these activities into one role-based system and uses email as an additional action channel.

The second motivation is explainability. TF-IDF and cosine similarity do not capture meaning as deeply as modern embedding models, but their terms and weights can be inspected. This makes them suitable for an academic baseline in which the scoring process, limitations, and effect of feedback must be demonstrated clearly.

The third motivation is controlled adaptation and automation. AutoFit places policy checks between a score and an action, considering consent, thresholds, previous interactions, cooldown, quota, time zone, and quiet hours. Rocchio feedback then provides a transparent way to adjust later rankings from explicit positive and negative judgments without overwriting the original representation.

Related work includes classical lexical retrieval, learned resume-job representations, explainable recommendation, and human-centered governance. Lexical methods provide inspectable weights and matching reasons, while recent dense approaches improve semantic representation but require stronger data and evaluation assumptions [6]-[8]. Research on algorithmic hiring and explainable recommendation also shows that model performance, fairness, user control, and

explanation quality must be evaluated separately [9], [10], [14]. CareerFit addresses a narrower integration gap by combining portal workflows, explainable scoring, feedback learning, policy-controlled actions, and audit records in one academic prototype.

Research Objectives

Overall Objective

The overall objective of this thesis is to design, implement, and evaluate a web-based Human-in-the-Loop recruitment automation platform for the IT domain. The platform is intended to support CV–job description evaluation, personalized job recommendation, feedback-based adaptation, and policy-driven actions while preserving user control, explainability, security, and auditability.

Specific Objectives

To achieve the overall objective, the thesis defines the following specific objectives:

- Design a role-based web platform for guests, candidates, recruiters, and administrators, with appropriate public and authenticated workflows.
- Provide candidate profile and CV management, including document upload, form-based entry, text extraction, OCR fallback, validation, and quality feedback.
- Implement separate CV–JD matching and candidate-profile recommendation workflows using a shared text-normalization, TF-IDF, and cosine-similarity pipeline.
- Present normalized scores, relevance labels, matching reasons, validation signals, and stable ranking behavior to support interpretation by candidates and recruiters.
- Integrate explicit user feedback and apply the Rocchio relevance-feedback method to update learned representations and recompute rankings reproducibly.
- Implement AutoFit policies that convert matching results into controlled outcomes according to consent, thresholds, interaction state, quota, cooldown, time-zone, and quiet-hour rules.
- Support expiring email-action links with hashed token storage, a non-mutating confirmation page, POST execution, action-state tracking, and audit logging.
- Evaluate the algorithmic behavior and system workflows using reproducible experiments, automated tests, scripted acceptance scenarios, and security and reliability checks appropriate to the project scope.

Research Subjects and Scope

Research Subject

The research subject is the IT recruitment workflow supported by a web platform, including CV and Job-description text, lexical ranking and recommendation, explicit feedback signals, policy-controlled automation, and auditable user and system actions.

Research Scope

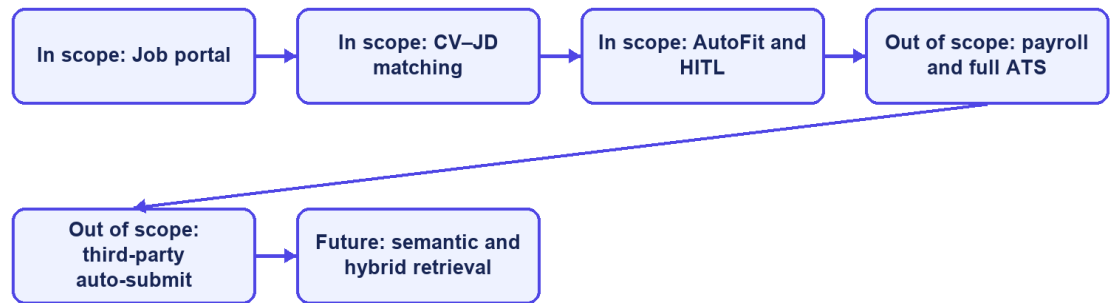
This thesis focuses on an academic prototype for recruitment in the information technology domain. Its functional scope includes a public job portal; authenticated candidate, recruiter, and administrator workspaces; CV and profile management; job and employer management; CV–JD matching; personalized job recommendation; application workflows; explicit relevance feedback; AutoFit policy configuration; notifications and actionable email; analytics; and audit records.

The data-processing scope includes extraction of text from supported CV documents, OCR fallback for image-based content where the configured runtime permits it, hard and soft validation, bilingual-oriented text normalization, TF-IDF vectorization, cosine-similarity scoring, score normalization, relevance labeling, and Rocchio-based feedback updates. PostgreSQL is used as the primary data store, and database evolution is managed through Flyway migrations. The implemented system follows a client–server architecture with a React and TypeScript frontend and a Spring Boot backend exposing REST APIs.

The thesis does not attempt to implement a complete enterprise applicant tracking system. Interview scheduling, offer management, payroll, and full employee lifecycle management are outside the core scope. The system does not automatically submit applications to external third-party job boards, and it does not use a large language model to make unrestricted recruitment decisions. Distributed microservices, large-scale message brokers, and semantic embedding models may be discussed as future extensions but must not be presented as implemented core components unless they are added and verified before the final report is submitted.

The evaluation scope is limited. Controlled or synthetic datasets can show whether Rocchio changes rankings in the expected direction, but they cannot prove recruitment quality in a real production setting. The source and purpose of scraped, seeded, or synthetic data must be stated. Claims about usability, performance, security, or production readiness are limited to the test environment, participants, procedures, and evidence available during the final evaluation.

Thesis scope boundary



CareerFit IT AutoPilot — thesis diagram

Figure 1.2. Scope boundary of the CareerFit thesis

Research Methods and Content

Research Methods

This thesis follows an applied software-engineering research approach. The work combines requirement analysis, architecture and data design, incremental implementation, a controlled algorithm experiment, automated backend and browser testing, runtime observation, and threats-to-validity analysis. Implementation claims are checked against source code, migrations, API contracts, logs, test reports, and generated evaluation artifacts.

Research Content

The research models CareerFit as a Perception-Decision-Action-Learning-Audit workflow. It first identifies recruitment roles, requirements, and control boundaries; then designs and implements the data, matching, recommendation, feedback, automation, email action, and audit modules. Finally, the study evaluates algorithm behavior, software correctness, browser workflows, authorization, runtime health, and local latency within the stated experimental scope.

Key Contributions

The principal contribution of this thesis is the design and implementation of an integrated, controlled recruitment workflow rather than the invention of a new machine-learning model. The contributions are summarized as follows:

- An integrated recruitment platform that combines a job portal, CV–JD matching, personalized recommendation, application management, and role-specific operational workspaces.
- A transparent text-scoring pipeline using TF-IDF and cosine similarity, with normalized results, labels, reasons, validation signals, and explicit separation between matching and recommendation objectives.
- A feedback-learning workflow based on the Rocchio method, designed to update learned representations from explicit feedback and to support reproducible recomputation rather than uncontrolled cumulative modification.
- A policy-driven AutoFit layer that separates relevance scoring from business actions and applies Human-in-the-Loop controls, user consent, operational constraints, and previous interaction states.
- An actionable email mechanism with expiring, hashed tokens, a non-mutating GET confirmation page, POST redemption, and action-state tracking.
- A traceable evaluation structure that distinguishes algorithm experiments from functional, integration, acceptance, performance, reliability, and security evidence, while explicitly documenting threats to validity.

Thesis objective	Implemented contribution	Evaluation evidence
Analyze IT recruitment workflows	Role-specific requirements and Human-in-the-Loop boundaries	Use cases and traceability matrix
Implement CV–JD matching	TF-IDF, cosine scoring, explanations, and persisted Matchings	Algorithm benchmark and backend tests
Learn from feedback	Rocchio updates followed by scheduled recomputation	Baseline-versus-learned ranking metrics
Support controlled automation	AutoFit policy, quota, cooldown, notification, and audit controls	Security tests and P0 E2E flows
Deliver an integrated platform	Spring Boot, React/TypeScript, PostgreSQL, and Flyway	Build, runtime health, and browser evidence

Table 1.1. Mapping of objectives, contributions, and evidence

Thesis Structure

The report contains an Introduction, four chapters, and a Conclusion. Chapter 1 defines the problem, stakeholders, requirements, and use cases. Chapter 2 presents the theoretical background and solution design. Chapter 3 explains the implementation. Chapter 4 presents the evaluation method, evidence, and threats to validity. The Conclusion discusses achievements, limitations, and future work.

MAIN CONTENT

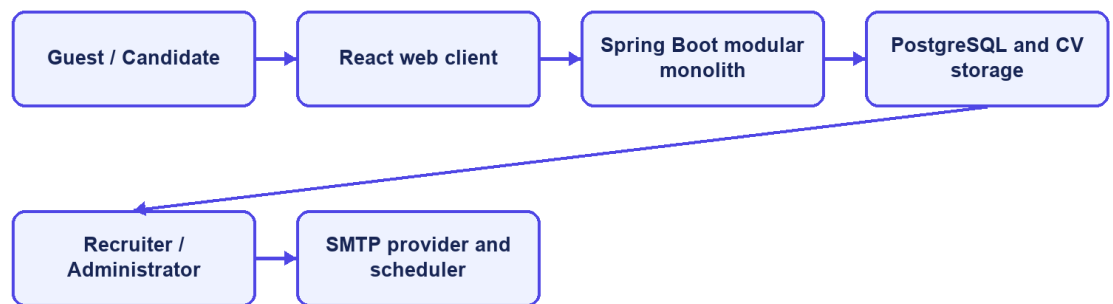
CHAPTER 1. PROBLEM DESCRIPTION AND REQUIREMENTS

1.1 System Analysis Context

CareerFit is designed as a role-based system for IT recruitment. Its boundary includes the React frontend, Spring Boot backend, PostgreSQL database, CV storage, background scheduler, and optional email provider. External job boards, interview scheduling, offers, payroll, and unrestricted automated hiring remain outside the implemented scope.

The central design distinction is between evidence, business state, and action. A matching record contains evidence about the similarity between one CV and one job. An application record represents a candidate's participation in a recruitment process. An automation policy represents consent and operational preferences. These concepts must remain separate: a high similarity score must not silently become an application, and an application status must not be interpreted as a relevance label.

System context



CareerFit IT AutoPilot — thesis diagram

Figure 1.3. CareerFit system context

1.2 Stakeholders and Actors

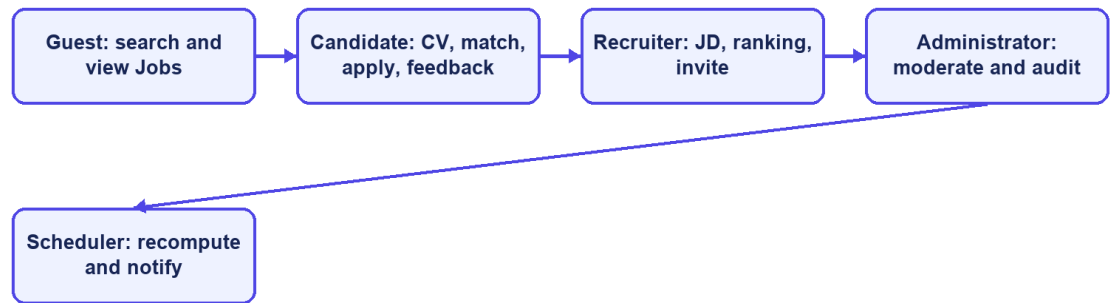
CareerFit serves four interactive roles and two supporting actors. Guest is a frontend state rather than a persisted account role. Candidate, Recruiter, and Administrator are persisted roles in `user_account`. Background processing and the mail provider are non-human actors that execute scheduled work and deliver messages.

Table 1.2. Actors and responsibilities

Actor	Primary responsibilities	Access boundary
Guest	Browse public jobs, search suggestions, job details, featured employers, and public market analytics	Public GET operations only; authentication is required for personalized data or application actions
Candidate	Maintain profile, CVs, and portfolio; view matching and recommendation results; apply or withdraw; provide feedback; configure personal automation	Candidate-owned resources and candidate APIs
Recruiter	Maintain employer profile and jobs; inspect applicants and discovered candidates; invite candidates; update application status; view recruiter analytics	Recruiter APIs and jobs owned by the authenticated recruiter
Administrator	Monitor system summaries, users, jobs, email actions/tokens, and audit records; moderate jobs and trigger controlled maintenance operations	Administrative APIs only
Background scheduler	Recompute stale matches, send digests and notifications, clean expired email actions, and execute auto-apply scans	Internal service calls under system control
Mail provider	Deliver passwordless or notification email generated by the backend	SMTP boundary; it does not own CareerFit business state

Stakeholder needs sometimes conflict. Candidates want relevant opportunities, privacy, and control over automatic applications. Recruiters want efficient prioritization without losing access to applicants who do not rank highly. Administrators need visibility and recoverability without unrestricted exposure of CV content. The design therefore combines role authorization, ownership checks, policy settings, validation, explicit statuses, and audit records rather than relying on a single ranking score.

Role-oriented use cases



CareerFit IT AutoPilot — thesis diagram

Figure 1.4. CareerFit use-case overview

1.3 Functional Requirements

The detailed Software Requirements Specification contains individual requirement identifiers. For the thesis, they are consolidated into functional groups that map to implemented modules and observable workflows.

Table 1.3. Consolidated functional requirements

Group	Required capabilities	Main actor
Authentication and account	Registration, password login, passwordless verification, current-account lookup, role enforcement, account settings	Candidate, Recruiter, Administrator
Public job portal	Search, suggestions, filters, job details, featured employers, employer details, and public analytics	Guest and authenticated users
Candidate profile and CV	Profile and portfolio maintenance, multiple CVs, default CV selection, document upload, manual creation, status tracking, validation, extraction, and OCR fallback	Candidate
Job and employer management	Employer profile management; job creation, update, status change, deletion, search, and CSV export	Recruiter
Matching and recommendation	CV–JD scoring, labels, reasons, candidate match cards, job ranking, profile-based recommendations, and similar jobs	Candidate and Recruiter

Group	Required capabilities	Main actor
Application workflow	Manual apply, candidate application history, withdrawal, recruiter applicant view, invitation, and status update	Candidate and Recruiter
Feedback learning	Record explicit feedback from web or email, update learned representations with Rocchio, and mark affected matches for recomputation	Candidate
Automation and notification	Policy retrieval/update, pause/resume, run-now, scheduled auto-apply, digest, high-match email, quota, cooldown, and delivery logging	Authenticated user and Scheduler
Analytics and administration	Market, candidate, recruiter, and system summaries; event tracking; user/job moderation; audit and email-token monitoring	Candidate, Recruiter, Administrator

The implementation exposes these groups through REST-oriented endpoints. Examples include `/api/jobs/search`, `/api/cv/upload`, `/api/matches/me/cards`, `/api/recommendations/jobs`, `/api/applications`, `/api/recruiter/jobs/{jobId}/candidates`, `/api/automation/policy`, and `/api/admin/audit-logs`. Chapter 3 describes endpoint behavior in detail. This chapter uses endpoint groups only to establish functional boundaries and traceability.

1.4 Non-Functional Requirements

Non-functional requirements constrain how the workflows must operate. They are expressed as design targets; Chapter 4 must distinguish targets from properties verified by measurement.

Table 1.4. Non-functional requirements and design responses

Quality attribute	Requirement	Design response
Security	Protected resources must require authenticated and authorized access	Stateless Spring Security filter chain, JWT processing, role rules, ownership checks, BCrypt password hashing, CORS allow-list, CSP, and HSTS configuration
Privacy	CV and account data must not be public or unnecessarily logged	Candidate-scoped APIs, local protected storage boundary, selective DTOs, and audit metadata rather than full document content

Quality attribute	Requirement	Design response
Reliability	Repeated or failed operations should not create inconsistent domain state	Database constraints, transactions, unique keys, optimistic version columns, idempotency checks, explicit statuses, and retry-aware background work
Performance	Public queries and ranking views should remain responsive for the academic dataset	Database indexes, pagination, stored vectors, background scoring, and bounded result sets
Maintainability	Domain changes should remain localized and database evolution reproducible	Modular package structure, controller–service–repository separation, DTOs, configuration properties, Flyway migrations, and automated tests
Usability	Users must understand loading, errors, empty results, score context, and available actions	Role-specific pages, normalized labels and reasons, validation signals, and explicit action controls
Auditability	Significant automated or administrative events should be reconstructable	audit_log, notification delivery records, email-action state, timestamps, actor/source metadata, and administrative views
Explainability	The system should expose evidence without presenting a score as a hiring probability	Lexical term/skill reasons, separate labels and application states, policy settings, and documented limitations

Security and auditability are defense-in-depth properties. A role rule at the URL layer does not replace ownership verification in a service, and an audit row does not make an unsafe action safe. Similarly, a database index is a performance design decision, not proof of latency under load. The evaluation chapter must test each critical claim in the environment in which it is reported.

1.5 Use-Case Analysis

1.5.1 Candidate Uploads a CV and Receives Matches

Table 1.5. Use case - Candidate uploads a CV and receives matches

Field	Description
Use-case ID	UC-01
Primary actor	Candidate
Preconditions	The Candidate account is active and authenticated.
Trigger	The Candidate uploads a supported document or submits manual CV data.
Main flow	Validate the source; store the CV; extract and normalize text; derive terms and skills; create or update the vector; score eligible active Jobs; persist unique CV-Job Matching records; return ordered match cards.
Alternative/exception flows	Reject unsupported or oversized files and invalid fields; record OCR or extraction failure; allow soft quality warnings; return an empty result when no active Job is eligible.